BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

PHYSICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The Davisson and Germer experiment confirmed:

(A) Wave nature of particles
(B) Polarization
(C) Compton effect
(D) Photoelectric effect

Q2. The frequency of an inductor of 1.0 H have a reactance of 500 Ω is:

(A) 70 Hz
(B) 80 Hz
(C) 90 Hz
(D) 100 Hz

Q3. Shunt resistance of an ammeter is:

(A) $Rs = (I - Ig) / Ig \times Rg$
(B) $Rs = (I - Ig)\, Rg$
(C) $Rs = Ig / Rg$
(D) $Rs = Ig / (I - Ig) \times Rg$

Q4. In a certain circuit, the transistor has a collector current of 10 mA and a base current of 40μA. The current gain of the transistor is:

(A) 250
(B) 300
(C) 350
(D) 450

Q5. The A.M. transmission frequencies range is from:

(A) 540 μHz to 1600 μHz
(B) 540 KHz to 1600 KHz
(C) 540 MHz to 1600 MHz
(D) 540 Hz to 1600 Hz

Q6. In case of capacitor, formula for energy density is given by:

(A) ½ εr ε0 E^2
(B) ½ εr ε0 E^2 Ad
(C) ½ εr E^2
(D) ½ ε0 E^2

Q7. The dead time of the G.M. counter is:

(A) ~ 10^{-8}s
(B) ~ 10^{-6}s
(C) ~ 10^{-4}s
(D) ~ 10^{-3}s

Q8. One henry is equal to:

(A) $VS^{-1}A^{-1}$
(B) VSA^{-1}
(C) $VS^{-1}A$
(D) VSA

Q9. When platinum wire is heated, it appears white at about:

(A) 500°C
(B) 900°C
(C) 1100°C
(D) 1600°C

Q10. The statement that the flux through any closed surface is 1/ε0 times the total charge enclosed in it was given by:

(A) Plank
(B) Maxwell
(C) Faraday
(D) Gauss

Q11. The curie temperature for iron is about:

(A) 75°C
(B) 750°C
(C) 7500°C
(D) 75000°C

Q12. The device which converts heat energy into electrical energy is:

(A) Thermocouple
(B) Solar cell
(C) Electric generator
(D) Cell

Q13. An atom can reside in metastable state for:

(A) ~ 10^{-6}s
(B) ~ 10^{-5}s
(C) ~ 10^{-7}s
(D) ~ 10^{-3}s

Q14. Output resistance of an operational amplifier is:

(A) A few ohms
(B) Mili ohms
(C) Mega ohms
(D) Micro ohms

Q15. One twelveth the mass of carbon atom is:

(A) 1 μ = 1.6606×10^{-31}kg
(B) 1 μ = 1.6606×10^{-27}kg
(C) 1 μ = 1.6606×10^{-15}kg
(D) 1 μ = 1.6606×10^{-6}kg

Q16. A device which converts electrical energy into mechanical energy is called:

(A) Transformer
(B) D.C. generator
(C) A.C. generator
(D) D.C. motor

Q17. Torque produces in a current carrying coil when it is placed in:

(A) Gravitational field
(B) Electric field
(C) Magnetic field
(D) Nuclear field

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) How can you identify that which plate of a capacitor is positively charged?

(ii) Electric lines of force never cross. Why?

(iii) Describe the effect of medium between two charges upon the Coulomb's force.

(iv) What was the result of Millikan oil drop experiment?

(v) How can a current loop be used to determine the presence of a magnetic field in a given region of space?

(vi) How lamp and scale arrangement is used to observe angle of deflection in galvanometer?

(vii) What is digital multimeter? What is its advantage?

(viii) Why the voltmeter should have a very high resistance?

(ix) A particle which produces more ionization is less penetrating. Why?

(x) What information is revealed by the length and shape of the tracks of an incident particle in Wilson Cloud Chamber?

(xi) Why are thermal reactors called “thermal”? How do they differ from fast reactors?

(xii) Define mass defect and binding energy.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) How does the motion of electrons in a n-type substance differ from motion of holes in a p-type substance?

(ii) How does electron-hole pairs produce current in a photo-voltaic cell?

(iii) Explain with the circuit diagram, how a transistor acts as an ON switch?

(iv) Why orderliness of domains of a ferromagnetic material is not preserved at high temperature? Explain.

(v) How many crystal systems exist based on geometrical arrangement? Draw schematics of any two systems.

(vi) What is meant by hysteresis loss? How is it used in construction of a transformer?

(vii) How many times per second will an incandescent lamp reach maximum brilliance when connected to a 50 Hz source?

(viii) In an RL circuit will the current lag or lead the voltage? Illustrate your answer by a vector diagram.

(ix) In an ideal parallel resonant circuit, why source current is zero at resonant frequency?

(x) What do you mean by drift velocity? What is the effect on the drift velocity by increasing the potential difference?

(xi) State Kirchhoff's first rule. Interpret it as law of conservation of electric charge.

(xii) Differentiate between ohmic and non-ohmic devices. Give one example of each.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) How would you position a flat loop of wire in a changing magnetic field so there is no emf induced in the loop?

(ii) Can a D.C. motor be turned into a D.C. generator? What changes are required be done?

(iii) What is back emf effect in motor?

(iv) When does light behave as a wave when does it behave as a particle?

(v) What happens to total radiations from black body if its absolute temperature is doubled?

(vi) What is general theory of relativity?

(vii) State and write formula of Compton effect.

(viii) Is energy conserved when an atom emits a photon of light?

(ix) Define ionization potential arid excitation potential.

> Note: SECTION – II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [5] Define electric potential difference and absolute electric potential. Also derive the relation between electric field and electric potential gradient such that $\bar{E} = \Delta V / \Delta r$

(b) [3] The resistance of an iron wire at 0°C is $1\times10^{4}\ \Omega$. What is the resistance at 500°C, if the temperature coefficient of resistance of iron is $5.2\times10^{-3}K^{-1}$?

Q6.

(a) [5] Derive the relation for energy stored in an inductor in terms of magnetic field.

(b) [3] A power line 10m high carries a current of 200A. Find the magnetic field of the wire at the ground.

Q7.

(a) [5] What is rectification? Discuss the phenomenon of semiconductor diode as full wave rectifier.

(b) [3] Find the value of current and inductive reactance when A.C voltage of 200 V at 50 Hz is passed through an inductor of 10 H.

Q8.

(a) [5] Define and explain photoelectric effect. Also explain photoelectric effect on the basis of quantum theory.

(b) [3] A 1.25 cm diameter cylinder is subjected to a load of 2500 kg. Calculate the stress on the bar in mega pascals.

Q9.

(a) [5] Explain experimental arrangement for the production of X-rays. What are characteristic X-rays? Discuss spectrum of X-rays.

(b) [3] Calculate the energy (in Mev) released in the reaction $^{2}_{1}H + {}^{3}_{1}H \longrightarrow {}^{4}_{2}He + {}^{1}_{0}n$ Mass of $^{2}_{1}H$, 2.014102u, $^{3}_{1}H$, 3.01605u, $^{1}_{0}n$, 1.008655u and $^{4}_{2}He$, 4.002603u

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The rest mass of photon is:

(A) Zero
(B) m_0
(C) Infinite
(D) 9.1×10^{-31} kg

Q2. Peak to peak value of A.C. is equal to:

(A) V_0 (B) $2V_0$

(C) $V_0/2$ (D) V_0^2

Q3. Voltmeter is a device like:

(A) High resistance galvanometer (B) Low resistance galvanometer

(C) Ammeter (D) Zero resistance galvanometer

Q4. The width of depletion region of pn junction depends upon which factor:

(A) Biasing conditions (B) Doping density

(C) Both A and B (D) None of these

Q5. Which of the following does not vary by varying the frequency of A.C.:

(A) Resistance (B) Capacitive reactance

(C) Inductive reactance (D) All of these

Q6. Electric field and electric potential are related by the relation:

(A) $E = \Delta V r$ (B) $E = -\Delta V/\Delta r$

(C) $E = K\, q/r^2$ (D) $E = -\Delta r/\Delta V$

Q7. Which of the following has greater ionizing power:

(A) α-rays (B) β-rays

(C) γ-rays (D) Neutrons

Q8. Mutual induction has practical role in the working of:

(A) Galvanometer (B) Ammeter

(C) A.C. generator (D) Transformer

Q9. When a platinum wire is heated it becomes white at about:

(A) 500°C (B) 1600°C

(C) 2000°C (D) 2500°C

Q10. Three capacitors each of capacitance 2μF are connected in parallel, their equivalent capacitance will be:

(A) 2 μF (B) 4 μF

(C) 6 μF (D) 8 μF

Q11. Above curie temperature, iron becomes:

(A) Ferromagnetic (B) Diamagnetic

(C) Insulator (D) Paramagnetic

Q12. The SI unit for electric power is:

(A) Horse power
(B) Watt
(C) K Wh
(D) JS

Q13. X-rays can be deflected by:

(A) Electric field
(B) Magnetic field
(C) Both Electric and magnetic
(D) Cannot be deflected by fields

Q14. A diode characteristic curve is a graph between:

(A) Current and time
(B) Voltage and time
(C) Voltage and current
(D) Current and resistance

Q15. After two half life, the fraction of the radioactive sample remain undecayed is:

(A) N_0
(B) $N_0/2$
(C) $N_0/4$
(D) $N_0/6$

Q16. Which of the following converts electrical energy into mechanical energy:

(A) A.C. generator
(B) Heat engine
(C) Transformer
(D) Motor

Q17. The range of ammeter can be increased, if the value of shunt resistance is:

(A) Decreased
(B) Increased
(C) Doubled
(D) Remains Constant

Fssay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) If a point charge q of mass m is released in a non-uniform electric field with field lines pointing in the same direction, will it make a rectilinear motion?

(ii) How can you identify that which plate of a capacitor is positively charged?

(iii) Is it possible to orient a current loop in a uniform magnetic field such that the loop will not tend to rotate? Explain.

(iv) Suppose that a charge q is moving in a uniform magnetic field with velocity 'v'. Why is there no work done by magnetic force on the charge q?

(v) How can radioactivity help in the treatment of cancer?

(vi) A particle which produce more ionization is less penetrating. Why?

(vii) What is the working of inkjet printer?

(viii) How mathematically we calculate electric intensity inside hollow charged sphere?

(ix) Explain the extension of right hand rule.

(x) What is pivoted type galvanometer? Give its working.

(xi) Compare α, β, γ radiation on the basis of: (a) Range in air. (b) Speed.

(xii) What is nuclear fission? Give example.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Do bends in a wire affect its electrical resistance? Explain.

(ii) What is meant by Wheatstone bridge? How can it be used to determine an unknown resistance?

(iii) Define tolerance. Give one example.

(iv) What is meant by para and diamagnetic substance? Give one example of each.

(v) Differentiate between ductile and brittle substances.

(vi) Write down the resistivities of conductors, insulators and semiconductors.

(vii) How does the motion of an electron in a n-type substance differ from the motion of holes in a p-type substance?

(viii) Why ordinary silicon diodes do not emit light?

(ix) Define rectification. Draw a circuit diagram for half wave rectification.

(x) What is meant by A.M. and F.M.?

(xi) What is choke? Why is it better as compared to a resistor to limit the current?

(xii) Write down any two properties of series resonance circuit.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Write four methods to change magnetic flux.

(ii) Is it possible to change both the area of the loop and the magnetic field passing through the loop and still not have an induced emf in the loop?

(iii) Four unmarked wires emerge from a transformer. What steps would you take to determine the turns ratio?

(iv) Why wave properties are predominate in radio wave? Explain.

(v) What is the energy of a photon of 400 nm?

(vi) We do not notice the de Broglie wavelength for a pitched cricket ball. Explain Why?

(vii) Can pair production take place in vacuum? Explain.

(viii) What are advantages of laser over ordinary light?

(ix) Using uncertainty principle, $\Delta x \Delta p = h/2\pi$. If diameter of nucleus is of the order of 10^{-14}m, prove that electron cannot reside the nucleus.

> Note: SECTION – II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [5] State and explain Kirchhoff's 2nd rule.

(b) [3] Two point charges $q_1 = -1\times10^{-6}$C and $q_2 = 4\times10^{-6}$C, are separated by a distance of 3.0 m. Find and justify the zero-field location.

Q6.

(a) [5] State and explain Lenz's law. Explain how Lenz's law is also a statement of law of conservation of energy.

(b) [3] How fast must a proton move in a magnetic field of 2.50×10^{-3}T such that the magnetic force is equal to its weight?

Q7.

(a) [5] How can we use an operational amplifier as a comparator? Discuss working of operational amplifier as night switch in comparator mode.

(b) [3] What is the resonant frequency of a circuit which includes a coil of inductance 2.5 H and a capacitance 40μF?

Q8.

(a) [5] Discuss the mechanism of electrical conduction by holes and electrons in a pure semi-conductor element.

(b) [3] Yellow light of 577 nm wavelength is incident on a cesium surface. The stopping voltage is found to be 0.25 V. Find (i) the maximum K.E. of the photoelectrons. (ii) the work function of cesium.

Q9.

(a) [5] Derive an expression for the quantized radii and energies of the orbits of hydrogen atom.

(b) [3] The half-life of $^{91}_{18}$Sr is 9.70 hours. Find its decay constant.